

|           |                                                                                                                                                  |           |
|-----------|--------------------------------------------------------------------------------------------------------------------------------------------------|-----------|
|           |                                                                                                                                                  |           |
| 5(a)      | energy per unit charge                                                                                                                           | <b>B1</b> |
|           | energy transferred by source driving charge around the complete circuit<br><b>or</b><br>energy transferred from other forms to electrical energy | <b>B1</b> |
| 5(b)      | there is a p.d. across the internal resistance/ $r$                                                                                              | <b>B1</b> |
|           | change in current/ $I$ results in a change in p.d. across the internal resistance                                                                | <b>B1</b> |
|           | $V = E - \text{p.d. across internal resistance}$<br><b>or</b><br>change in p.d. across $r$ causes a change in $V$ (as e.m.f. is constant)        | <b>B1</b> |
| 5(c)(i)   | $E = 7.4 \text{ V}$                                                                                                                              | <b>A1</b> |
| 5(c)(ii)  | maximum current = 0.92 A                                                                                                                         | <b>A1</b> |
| 5(c)(iii) | $r = E / I_{\text{MAX}}$ <b>or</b> (-)gradient                                                                                                   | <b>C1</b> |
|           | e.g. $r = 7.4 / 0.92$<br><br>$= 8.0 \Omega$                                                                                                      | <b>A1</b> |
| 5(d)      | straight line with negative gradient that is smaller in magnitude than the original line                                                         | <b>B1</b> |
|           | line which would have intercept on $V$ -axis below the original line                                                                             | <b>B1</b> |